

What have we been doing
this week?

MATH 213

What we have been doing this week:

Learning how to draw causal conclusions from experiments.

- Situation: Comparing a treatment group with a control group in an experiment. Say it looks like the treatment is having the desired effect within the context of the experiment. But what if the difference we are noticing between the groups is just a byproduct of the random assignment process? How can we determine causation?
- This week: Performing a randomization hypothesis test can help us to test the notion that the difference between groups is due to random assignment.

Recap: Hypothesis Testing Framework

- We start with a *null hypothesis* (H_0) which is the hypothesis of no effect.
- We also have an *alternative hypothesis* (H_A) that represents our research question, i.e. what we're testing for.
- We conduct a hypothesis test under the assumption that the null hypothesis is true, either via simulation (this week) or theoretical methods (perhaps later in the course).
- If the experimental test statistic (in our case this week: the test statistic was a difference of means or a difference of proportions) looks very unusual compared to the dotplot of statistics generated in the simulation under the null hypothesis, then we have evidence to suggest that the experimental results are incompatible with the null hypothesis. Then we can reject the null hypothesis.
- If the experimental test statistic looks to be compatible with the dotplot generated under the null hypothesis, it DOES NOT mean that the null hypothesis is true. It doesn't tell us much at all except that perhaps we need to gather more data!

Notation Interlude

We always use the population notation in our hypotheses. We *know* specifics about our sample notation; the whole point is we're trying to figure out information about the population.

	Sample	Population
Mean (from a numerical variable)	$\bar{x}$ or just say “x-bar”	μ (lowercase Greek letter mu)
Proportion (from a categorical variable)	p	$\hat{p}$ Or “p-hat”

Examples of null and alternative hypotheses

Experiment	Null Hypothesis, H ₀	Alternative Hypothesis H _a
Bank supervisors	The gender associated with the file had no effect on the supervisors' decision to promote or not. In symbols: $p_{\text{male}} - p_{\text{female}} = 0$	The gender associated with the file did have an effect: male files caused supervisors to promote. In symbols: $p_{\text{male}} - p_{\text{female}} > 0$
Weight Loss	The weight loss plan had no effect on the participants' weight loss. In symbols: $\mu_{\text{treatment}} - \mu_{\text{control}} = 0$	The weight loss plan did have an effect: the incentive plan caused people to lose weight. In symbols: $\mu_{\text{treatment}} - \mu_{\text{control}} > 0$

What is a p-value?

To judge how unusual the experimental result looks in comparison to the dotplot of statistics generated in the simulation under the null hypothesis, people often calculate a p-value.

What is a p-value?

A ***p-value*** is the probability of obtaining a test statistic as or more extreme than the experimental result under the assumption of the null hypothesis. The smaller the p-value, the more unlikely the result *if the null is true*.